1. Input

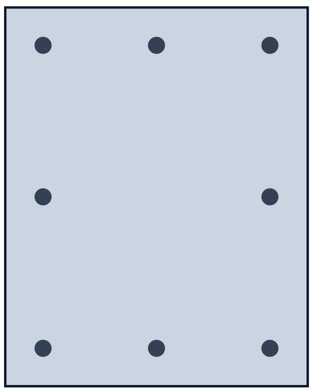
Section, materials and resistance basis exactly as analysed.

Identity

Project	**ACI-EB 01 - Rectangle 400 x 500 m.**	Section	**Benchmark 01 - Rectangle 400 x 50.**
Calculation profile	**ACI 318-19(22) — Whitney equivale.**	Mechanics	**Equivalent rectangular stress block**
Resistance profile	**ACI CODE-318-19 (Reapproved 202.**	Profile edition	**2019 (Reapproved 2022)**
Verification status	**draft**	Concrete model basis	**Code-default**
Method id	**aci-318-19-22-global-strength-reduct.**	Units	**mm · N · MPa; forces reported in kN,.**
Sign convention	**Compression positive; Mx = ΣF·(y-yc.**		

Section

Cross-section (mm, about the net concrete centroid) 8 bars



Section properties

Gross concrete area Ag	**200,000 mm^2**
Reinforcement area As	**3,200 mm^2**
Reinforcement ratio ρ	**1.600 %**
Bar count	**8**
Net centroid	**(0.0, 0.0) mm**
Bounding box	**400.0 × 500.0 mm**
Solids / holes	**1 / 0**
Perimeter	**1,800.0 mm**

Materials

Name	**Concrete C40**	Standard	**ACI318**
fck	**40.0 MPa**	Model	**aci-whitney-block**
εcu	**0.00300**	α	**0.8500**
Ec	**29,725 MPa**	Tension	**ignored**
Name	**Rebar fy 420**	Standard	**ACI318**
fy	**420.0 MPa**	Es	**200,000 MPa**
Model	**elastic-perfectly-plastic**	εy	**0.002100**
εu	**not declared**	Grades used	**1**

Design resistance basis

Format	**Global resultant strength-reduction f.**	Transverse reinforcement	**other**
ϕ compression (ties)	**0.650**	ϕ compression (spiral)	**0.750**
ϕ tension	**0.900**	εt,limit	**0.00510**
Transition rule	**yield-plus-strain**	Axial cap	**0.800 × factored compression pole**

Sampling and numerical evidence

Sampling mode	**fixed**	Active Design points	**777**
Active Design directions	**36**	Active Design stations	**27**
Active Nominal points	**902**	Active Nominal directions	**36**
Active Nominal stations	**27**	Direction tolerance	**0.000 %**
Worst interpolation error	**9.7950 %**	Tolerance reached	**NO**
Concrete discretization	**exact polygon clipping**	Strain domain	**concrete-pivot-ultimate**

Scope and limitations

This document reports a preview calculation. It is not an accepted design result and must not be released as a design report.

Resistance is reported in two stages: Nominal is the reference before the resistance treatment, Design is the usable resistance. Only the Design surface is compared with factored ULS demand.

Concrete compression is the code-equivalent rectangular block over a = β1·c, obtained by exact polygon clipping. There is no concrete integration mesh, and the stress between a and c is zero.

Overview interaction diagrams cut the active fixed 27-station, 10° direction surface. An arbitrary exact direction is calculated only when it is requested.

Reinforcement schedule

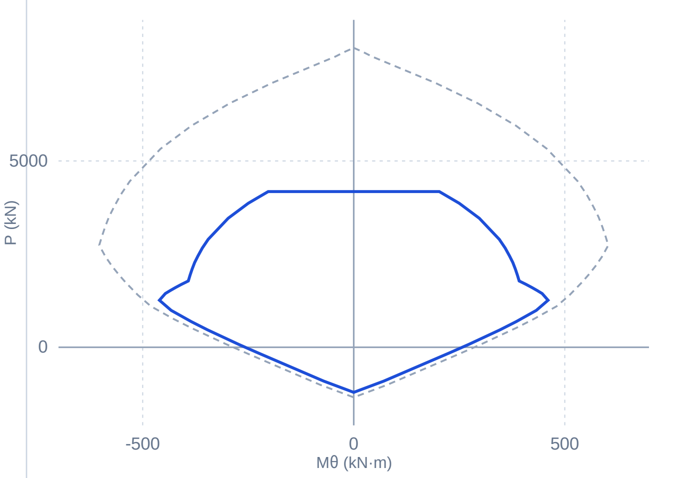
#	Bar id	Ø (mm)	X (mm)	Y (mm)	As (mm²)	Steel
1	1	22.6	-150.0	-200.0	400.0	Rebar fy 420
2	2	22.6	0.0	-200.0	400.0	Rebar fy 420
3	3	22.6	150.0	-200.0	400.0	Rebar fy 420
4	4	22.6	150.0	0.0	400.0	Rebar fy 420
5	5	22.6	150.0	200.0	400.0	Rebar fy 420
6	6	22.6	0.0	200.0	400.0	Rebar fy 420
7	7	22.6	-150.0	200.0	400.0	Rebar fy 420
8	8	22.6	-150.0	0.0	400.0	Rebar fy 420

2. Section capacity

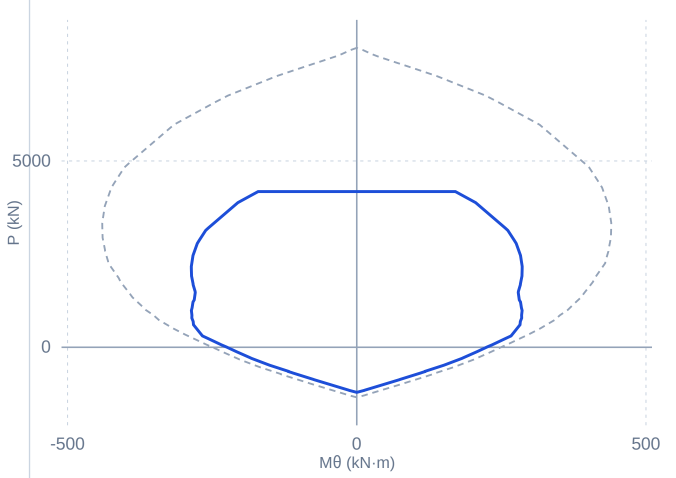
Nominal reference and Design resistance. Only Design is compared with factored demand.

Maximum design axial resistance applied at 4178.3 kN; it appears as the flat top of each diagram.

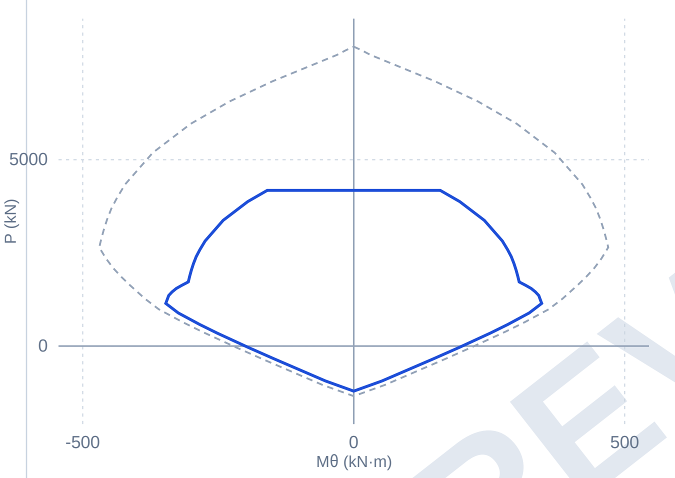
Fixed-grid plane P–Mθ at θ = 0.00° Design solid · Nominal dashed



Fixed-grid plane P–Mθ at θ = 51.25° Design solid · Nominal dashed



Fixed-grid plane P–Mθ at θ = 90.00° Design solid · Nominal dashed



3. Factored ULS demand

Governing check is the proportional 3D ray against the Design surface.

#	Combination	Pu (kN)	Mux (kN·m)	Muy (kN·m)	θL (°)	ϕ	Class	UR	Verdict
1	Audit ray 55% - NA 26.93 d.	661.2	97.5	121.5	51.3	0.694	transition	0.550	ADEQUATE
2	Cardinal audit 85% - NA 0 d.	2178.8	0.0	242.0	90.0	0.650	compression-co.	0.850	ADEQUATE
3	Cardinal audit 85% - NA 90 .	1606.2	330.7	0.0	0.0	0.650	compression-co.	0.850	ADEQUATE

UR is the governing proportional utilization: the factored demand vector scaled until it meets the Design surface. A fixed-axial ratio is reported per combination in its detail section and is a secondary diagnostic only.

4.1 Audit ray 55% - NA 26.93 deg, c/D 0.55

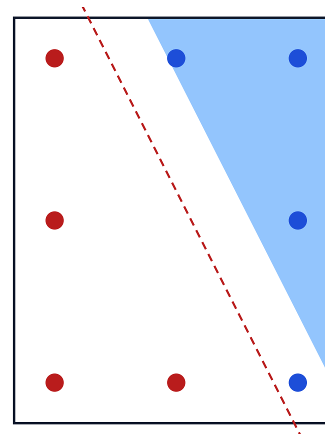
Pu = 661.2 kN · Mux = 97.5 kN·m · Muy = 121.5 kN·m · θL = 51.25°

ADEQUATE · utilization 0.5500 · fixed-P diagnostic 0.5491

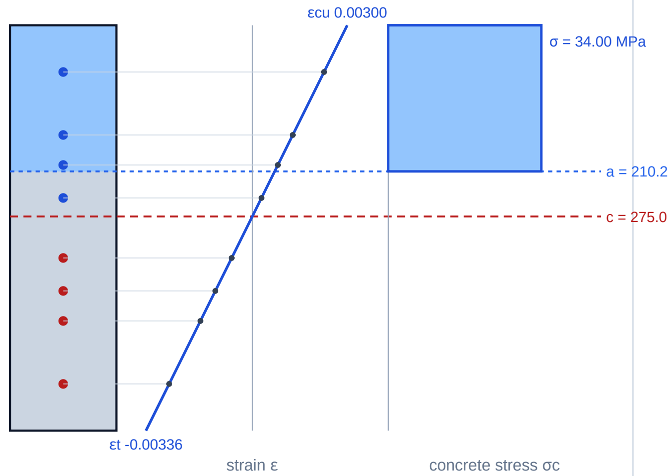
Factored ULS demand checked by the equivalent-block proportional-ray solver.

Section state

Transverse section — neutral axis and compression zone a=210



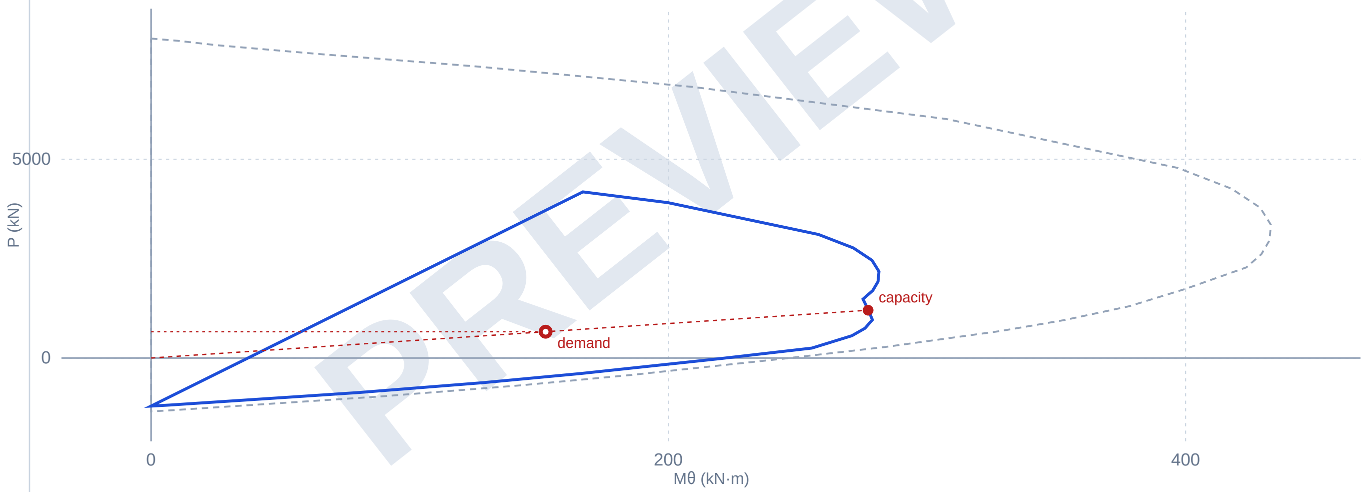
Longitudinal section — strain and stress over the depth Dθ=583



Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Exact direct meridian at the equilibrium strain direction

Direct P–Mβ at βeq = 63.071°, with the load point solid = fixed Design · dashed = fixed Nominal · points are projected on βeq



Concrete and resultants

Block area Ac,blk	**54,702.0 mm²**	Block centroid	**(121.4, 95.3) mm**
Block stress σblock	**34.000 MPa**	Concrete force Cc	**1,859.9 kN**
Concrete Mcx	**177.3 kN·m**	Concrete Mcy	**225.8 kN·m**
Nominal Pn	**1,732.5 kN**	Nominal Mnx	**255.6 kN·m**
Nominal Mny	**318.5 kN·m**	ϕ	**0.6940**
Design ϕPn	**1,202.3 kN**	Design ϕMnx	**177.4 kN·m**
Design ϕMny	**221.0 kN·m**		

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σs (MPa)	F (kN)	in block
1	-150.0	-200.0	400.0	515.8	-2.627e-3	-420.00	-168.00	no
2	0.0	-200.0	400.0	382.1	-1.168e-3	-233.70	-93.48	no
3	150.0	-200.0	400.0	248.4	2.904e-4	58.09	23.23	no
4	150.0	0.0	400.0	157.8	1.279e-3	255.71	88.68	yes

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σs (MPa)	F (kN)	in block
5	150.0	200.0	400.0	67.2	2.267e-3	420.00	154.40	yes
6	0.0	200.0	400.0	201.0	8.077e-4	161.55	51.02	yes
7	-150.0	200.0	400.0	334.7	-6.512e-4	-130.24	-52.10	no
8	-150.0	0.0	400.0	425.3	-1.639e-3	-327.86	-131.15	no

Solver and admissibility evidence

Neutral-axis angle θ	**26.9290°**	Neutral-axis depth c	**275.000 mm**
Block depth a	**210.179 mm**	β1	**0.7643**
Projected depth Dθ	**583.07 mm**	Controlling εt	**0.002627**
Controlling bar	**1**	Classification	**transition**
Axial cap governs	**no**	Component residual	**-5.82e-11**
Strain admissibility	**admissible**	Governing utilization	**0.5500**

4.2 Cardinal audit 85% - NA 0 deg, c/D 0.55

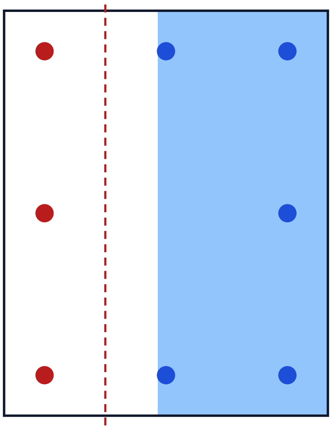
P_u = 2178.8 kN · M_{ux} = 0.0 kN·m · M_{uy} = 242.0 kN·m · θ_L = 90.00°

ADEQUATE · utilization 0.8500 · fixed-P diagnostic 0.8163

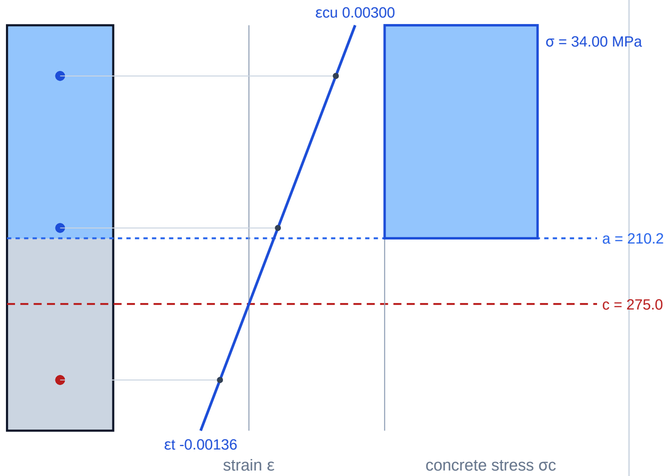
Factored ULS demand checked by the equivalent-block proportional-ray solver.

Section state

Transverse section — neutral axis and compression zone a=210



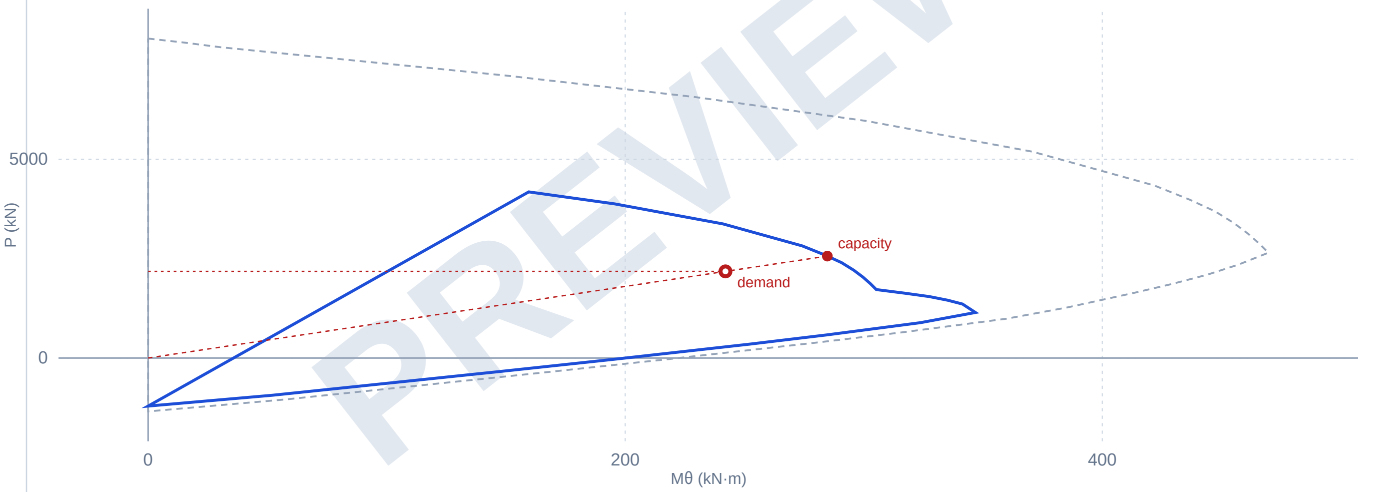
Longitudinal section — strain and stress over the depth Dθ=400



Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Exact direct meridian at the equilibrium strain direction

Direct P–Mβ at βeq = 90.000°, with the load point solid = fixed Design · dashed = fixed Nominal · points are projected on βeq



Concrete and resultants

Block area $A_{c,blk}$	**105,089.3 mm²**	Block centroid	**(94.9, -0.0) mm**
Block stress σ_{block}	**34.000 MPa**	Concrete force C_c	**3,573.0 kN**
Concrete M_{cx}	**-0.0 kN·m**	Concrete M_{cy}	**339.1 kN·m**
Nominal P_n	**3,943.6 kN**	Nominal M_{nx}	**-0.0 kN·m**
Nominal M_{ny}	**438.1 kN·m**	ϕ	**0.6500**
Design ϕP_n	**2,563.3 kN**	Design ϕM_{nx}	**-0.0 kN·m**
Design ϕM_{ny}	**284.7 kN·m**		

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σs (MPa)	F (kN)	in block
1	-150.0	-200.0	400.0	350.0	-8.182e-4	-163.64	-65.45	no
2	0.0	-200.0	400.0	200.0	8.182e-4	163.64	51.85	yes
3	150.0	-200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
4	150.0	0.0	400.0	50.0	2.455e-3	420.00	154.40	yes

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σs (MPa)	F (kN)	in block
5	150.0	200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
6	0.0	200.0	400.0	200.0	8.182e-4	163.64	51.85	yes
7	-150.0	200.0	400.0	350.0	-8.182e-4	-163.64	-65.45	no
8	-150.0	0.0	400.0	350.0	-8.182e-4	-163.64	-65.45	no

Solver and admissibility evidence

Neutral-axis angle θ	**0.0000°**	Neutral-axis depth c	**275.000 mm**
Block depth a	**210.179 mm**	β1	**0.7643**
Projected depth Dθ	**400.00 mm**	Controlling εt	**0.000818**
Controlling bar	**1**	Classification	**compression-controlled**
Axial cap governs	**no**	Component residual	**-1.75e-10**
Strain admissibility	**admissible**	Governing utilization	**0.8500**

4.3 Cardinal audit 85% - NA 90 deg, c/D 0.55

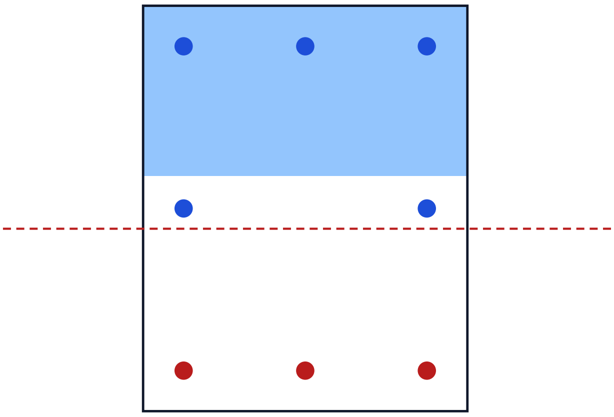
Pu = 1606.2 kN · Mux = 330.7 kN·m · Muy = 0.0 kN·m · θL = 0.00°

ADEQUATE · utilization 0.8500 · fixed-P diagnostic 0.7840

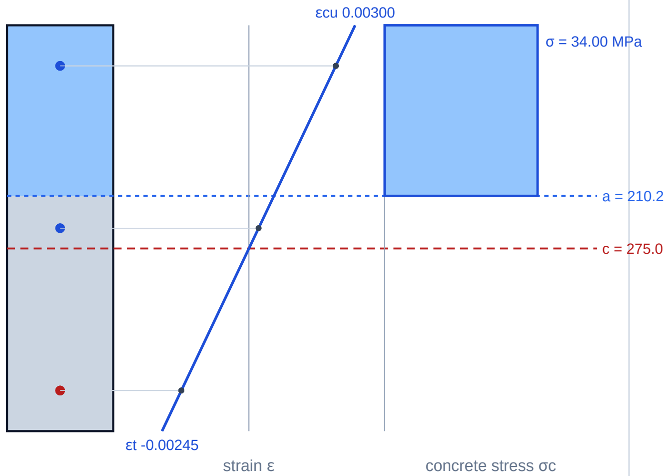
Factored ULS demand checked by the equivalent-block proportional-ray solver.

Section state

Transverse section — neutral axis and compression zone a=210



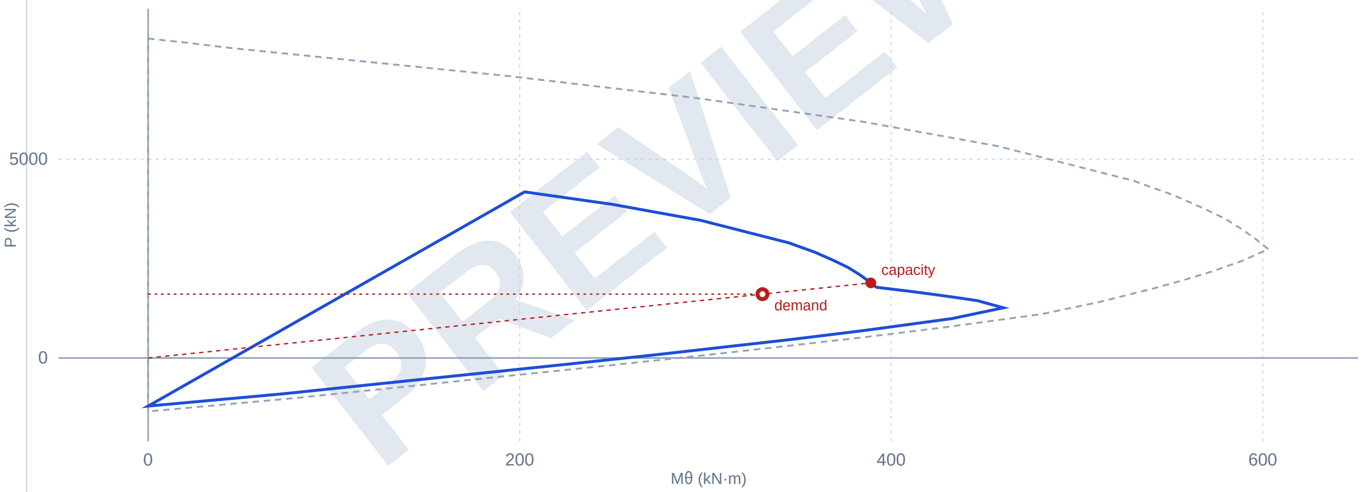
Longitudinal section — strain and stress over the depth Dθ=500



Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Exact direct meridian at the equilibrium strain direction

Direct P–Mβ at βeq = 0.000°, with the load point solid = fixed Design · dashed = fixed Nominal · points are projected on βeq



Concrete and resultants

Block area Ac,blk	**84,071.4 mm²**	Block centroid	**(0.0, 144.9) mm**
Block stress σblock	**34.000 MPa**	Concrete force Cc	**2,858.4 kN**
Concrete Mcx	**414.2 kN·m**	Concrete Mcy	**0.0 kN·m**
Nominal Pn	**2,907.1 kN**	Nominal Mnx	**598.5 kN·m**
Nominal Mny	**0.0 kN·m**	ϕ	**0.6500**
Design ϕPn	**1,889.6 kN**	Design ϕMnx	**389.0 kN·m**
Design ϕMny	**0.0 kN·m**		

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σs (MPa)	F (kN)	in block
1	-150.0	-200.0	400.0	450.0	-1.909e-3	-381.82	-152.73	no
2	0.0	-200.0	400.0	450.0	-1.909e-3	-381.82	-152.73	no
3	150.0	-200.0	400.0	450.0	-1.909e-3	-381.82	-152.73	no
4	150.0	0.0	400.0	250.0	2.727e-4	54.55	21.82	no

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σs (MPa)	F (kN)	in block
5	150.0	200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
6	0.0	200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
7	-150.0	200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
8	-150.0	0.0	400.0	250.0	2.727e-4	54.55	21.82	no

Solver and admissibility evidence

Neutral-axis angle θ **90.0000°**
Block depth a **210.179 mm**
Projected depth Dθ **500.00 mm**
Controlling bar **1**
Axial cap governs **no**
Strain admissibility **admissible**

Neutral-axis depth c **275.000 mm**
β1 **0.7643**
Controlling εt **0.001909**
Classification **compression-controlled**
Component residual **-7.28e-11**
Governing utilization **0.8500**